

RUPERTO PEDRO BONET CHAPLE

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Summary

Highly experienced mathematics educator and researcher with a Ph.D. in Computational Mechanics. Over 40 years of teaching combined with recent advancements in Python development, machine learning, and real-world data science projects.

Employment History

Dates (month/year) From To		Employer and/or institution in which employed	Position held, duties and responsibilities	Full/part time
Dec/2022	Now	Employer: Madison Reign agency Institution: Kingsdale Foundation School	Maths Tutor - Teach GCSE students with learning difficulties in a face-to-face setting. - Develop tutorial materials in their simplest forms for easy assimilation by students. - Encourage, support, and challenge students to meet their academic/math targets.	Part time
April/2021	Dec/2022	Self-employment,	Maths Tutor - Teach GCSE, A-Level and university students online and face-to-face. - Help get students prepared for tests and exams. - Note the performance of the students - Invigilate exams in different academies	Part Time
Oct/2020	April/2021	Worldwide University of London	Online Tutor Msc. Data Science Tutor university students in Calculus, Geometry, and Statistics with high achievement.	Part Time

			- Develop lectures via webinars on Calculus and Statistics. - Mark exams for the related courses.	
Sept/2015	Oct/2016	Technical University of Babahoyo (UTB-Ecuador)	Associate Professor: Senescyt Program Ateneo - Teach courses on differential and Integral Calculus, Descriptive and Inferential statistics, and Matlab for undergraduate students. - Teach a postgraduate course on Statistics Applied to the investigation - Organize, develop and manage statistical projects in Agriculture, Education, Commercial, and Business industries. - Participate as a Member of the Scientific Committee for the International Congress of Technology Trends, Babahoyo, Ecuador (2016). - Develop a research group based on the "Computational Methods for the Flooding Risk Management in the Babahoyo River Basin, Ecuador (2016-2018)" project. - Publish a scientific article: A parametric method for the shallow water equations (89-102). Journal of Magazine of the Sciences. ISSN 2528-8091. Vol1. No 4, October-December 2016	Full time
Feb/2014	Sept/2015	Funiber Editorial, Barcelona, Spain	Writer/Author - Author of the Calculus and Differential book (ISBN 978-84-9079-205-6). - Review a Finance Mathematics book.	Full time
Aug/2010	Feb/2014	National University of San Luis, Argentina	Associate Professor - Teach undergraduates courses in Engineering and Mathematics careers: Partial Differential Equations, Complex Variable Analysis, Fourier Analysis, Applied Mathematics (Aug/2010-Mar/2014), Basic Mathematics, Modeling and Numerical Methods, Series and Partial Differential Equations(Aug/2000-Mar/2001) - Manage the Magister Thesis entitled "The π number and its historical evolution" at Sarmiento University, Buenos Aires, Argentina (2012). - Deploy a learning project entitled Design of the Applied Mathematics course for undergraduate engineers (2013). - Research about Numerical	Full time

			Methods in Partial Differential Equations. Reference PROYCO 22 /F7300 Participate in Congress: • Speaker. Improved discrete non-local absorbing boundary condition for Helmholtz equation. Applications to unbounded Wave Problems. III MACI 2011. Vol. 3, 267-270. White Bay, Argentina (9-11 May 2011). • Speaker. Residual Discrete Non-Local Method for Waves Radiation and Scattering Problems. IVCLAM.Córdoba, Argentina. (August 6-10, 2012). • Publish an article entitled: "Refraction and Diffraction of Water Waves using finite elements with a DNL boundary condition". Ocean Eng. Journal 63, pages 77-89, 2013.	
Jul/2002	Mar/2008	Polytechnic University of Catalonia(UPC), Barcelona, Spain	Contracted Doctor: Ramon y Cajal Researcher • Teach undergraduate courses: Integral Calculus, Ordinary Differential Equations, Rational Mechanics for Civil Engineers, and Physical Phenomenon for Engineering. • Conduct the Research project Computational Methods for Transport Phenomena in rivers, estuaries and coasts. • Participate in the research project Turbulent Diffusion of marine contaminants, analysis and tracking by satellites (SAR/ASAR, IR) and use in predictive numerical modelling, FA-UPC, Spain, 2005-2008. • MEC. PIP N. ESP2005 – 07551, (Collaborator) • Visiting Professor/ Researcher • Benemérita Autonomous University of Puebla (BUAP, Mexico) Visiting Professor on April /2006 • Technical University of Delft (TU Delft-Netherlands) Visiting Researcher from /2/15/2006 to 3/15/2006 • Technical University of Dresden(TU Dresden-Germany) Visiting Researcher from 2/1/2006 to 2/10/2006 Participate in Congress:	Full time

			• Speaker: Outgoing Boundary Conditions on a Non-circular Open Boundary for Elliptic Harbor Wave Models. VII International Conference on Advances in Turbulence. Barcelona, Spain (21-23 November 2003) • Speaker: Comparison of Outgoing Boundary Conditions Strategies for Elliptic Harbor Waves Models. Eccomas 04. Jyvaskila Faculty of Engineering, Finland (2004). • Speaker: An Extended Elliptic Water Waves Model to Simulate Waves in Harbors. Marine 2005. Oslo, Norway (June 27-30, 2005). Published in the proceedings Computational Methods in Marine Engineering. Pages 455-46. ISBW: 84-95999-63-3. Organized by CIMNE/Det Norske Veritas • Speaker: Numerical Simulation of Wave Propagation in Toulon Bay. Waves 2005. Madrid, Spain, (July 3-7, 2005). NIPO163-05-009-006. Organized by CEDX/IACHR. • Speaker: Numerical Solution of an Unbounded Wave Guide Problem. IABEM 2006. Graz. Austria (July 10-12, 2006). Published on CD: ISBN: 3-902465-23-9. Organized by Graz University of Technology Publish the following articles: • Stabilized Element Methods for Transport Equations”, Technical Report, Barcelona, April 15, (2004). • Numerical Solution of Refraction-diffraction of Surface Waves, RIMNE, Vol. 21 (2), pages 143-168, (2005). ISSN: 0213-1315-in Spanish. • A local Absorbent Boundary Conditions to Berkhoff Equitation over a General Shape boundary, RIMNE, Vol. 21(1), pages 103-133, (2005). ISSN: 0213-1315- In Spanish. • Numerical Stabilization of Convection-diffusion-reaction Problems”, Technical Report 06-03 ISSN 1389-6520, Delft, pages 1-19, (2006). • Conduct magister thesis on “Numerical Computations of Irregular Waves in Hyeres” (2004)	
Sept/1982	Jul/1999	Higher Polytechnic Institute Jose A. Echeverria (ISPJAE-Cuba)	Associate Professor • Teach undergraduate engineering courses: Calculus, Mathematics Analysis, Descriptive Statistics, Basic Mathematics, Series and Partial Diff Equations. • Conduct the Degree thesis Waves in Harbours at Havana University, 1982. • Design the course Linear Algebra and its Applications to Engineering	Full time

			(1988). - Develop the learning project entitled Introduction the Active Methods in the Classroom (1990-1992).	
Sept/1979	Jul/1982	Ministry of Education-Cuba	Education Manager at National Level - Teach courses to professors and Methodological Managers of Secondary Schools on different subjects (1980-1982). - Supervise Mathematics education to Secondary school teachers (1979-1980). - Work in a National Comision of Planes and Programs related to Mathematical Education Author. [Text Book] Mathematics (1984). - Editorial People and Education. In Spanish. Cuba. Author. [Book].Course for primary school teachers. Methodological Guide to Study (1983). In Spanish. Cuba.	Full time

Education

Dates (month/year) From To		Institution of Higher Education	Subjects taken	Qualifications gained	Full/part time
Feb/1989	Feb/ 1989	ISPJAE-Cuba	Hydraulics for mathematicians	Excellent	40 hrs
June/1991	July/ 1991	ISPJAE-Cuba	Finite Element Methods	Excellent	48 hrs
Mar/1996	June/1996	Computational Methods Research Centre-National University of the Litoral (CIMEC-UNL, Argentina)	Epistemology	Distinguished	50 hrs
Mar/1996	Aug/1996	CIMEC-UNL, Argentina	Partial Differential Equations	Outstanding	60 hrs
Sept/1996	Dec/ 1996	CIMEC-UNL, Argentina	Introduction to Finite Elements	Outstanding	70 hrs
Mar/1997	Aug/1997	CIMEC-UNL, Argentina	Introduction to Mathematical Methods in Engineering	Pass	120 hrs
Mar/1997	Aug/1997	CIMEC-UNL, Argentina	Introduction to Scientifica Calculus with Parallel Computing	Distinguished	45 hrs
Mar/1997	Aug/ 1997	CIMEC-UNL, Argentina	Fluid Mechanics	Good	150 hrs

Jun/ 1998	Jun/1998	Faculty of Engineering and Water Sciences (FICH-UNL,Argentina)	The impact of technology on port engineering and operation and on the development of river routes	Pass	20 hrs
Feb/1995	Jul/1999	CIMEC-UNL, Argentina	Mathematical Models of Hydrodynamic Coastal Problems	Ph.D. (Good)	Full Time
Apr/2000	Jul/2002	CIMEC-UNL, Argentina	Numerical Models for rivers, estuaries and coastal hydrodynamic problems.	Post-Doctoral Scholarship	Full Time

Dates (month/year) From To		Institution of Higher Education	Subjects taken	Qualifications gained	Full/part time
Apr/2001	Jun/2001	International Center of Numerical Methods in Engineering-Polytechnic University of Catalonia, Spain (CIMNE-UPC,Spain)	Stabilization techniques for the transport equation based on the FIC method	Post-Doctoral scholarship	Full Time
Jul/2002	Jul/2002	UPC, Spain	Turbulent diffusion in the environment	Attendance	20 hrs
Sept/2003	Sept/2003	UPC, Spain	Turbulent flows of variable density	Attendance	20 hrs
May/2005	May/2005	UPC, Spain	Open boundary conditions to Helmholtz equation	Attendance	Seminar
Feb/2006	Feb/2006	Numerical Mathematical Institute of Dresden.TU-Dresden, Germany	Computational Methods to Transport Equations in Coastal Process	Attendance	Seminar
Feb/2006	Mar/2006	TU-Delft, Netherlands	Numerical Stabilization of Convection-diffusion-reaction Problems	Attendance	Full Time
Jul/2002	July/2007	UPC-Barcelona, Spain	Computational Methods for Transport Problems.	Post-Doctoral scholarship	Full Time
Feb/2014	Dec/2014	UPC-Barcelona,Spain	Varieties in Algebraic Geometry	Research stage	Part Time
May/2016	Sept/2016	Babahoyo Technical University	Applied Statistics to Scientific Research	Good	80 hrs
Jun/2019	Jun/2019	IBM (Coursera Certification)	Python for Data Science and AI	93.25%	25 hrs

Apr/2019	Apr/2019	IBM (Coursera Certification)	Applied AI with DL Advanced	96.88%	25 hrs
March/2020	March/2020	HSE University (Coursera Certification)	Natural Language Processing	93.01%	32 hrs
Oct/2020	Oct/2020	The State University of New York (Coursera Certification)	Practical Time Series Analysis	97.03%	25 hrs
Dec/2020	Dec/2020	DeepLearning AI (Coursera Certification)	Sequences.Time Series and Prediction	93.75%	22 hrs
Feb/2021	Feb/2021	HSE University (Coursera Certification)	Practical Reinforcement Learning	99.13%	26 hrs
Mar/2021	Mar/2021	New York Institute of Finance (Coursera Certification)	Using ML in Trading and Finance	100%	18 hrs

Membership

Member of Scientific Committee for the International Congress of Technology Trends, Babahoyo, Ecuador (2016)

Member of the Argentinean Association of Mathematics (UMA) 2011-2013

Active Participation in Spanish Society on Numerical Methods in Engineering(SEMNI), 2002-2008.

Member of the Argentinean Association of Computational Mechanics (AMCA),1995-2002:

Member of the Committee of the Mathematics And Computing Cuban Society (SCMC- Havana), 1993

Member of the Jury of the Fifth National Pedagogic Day, ISPEJV- Havana,1989

Member of the Jury of the National Olympic of Mathematics, Cuba,1982

Member of the Jury of the Fifth National Pedagogic Day, ISPEJV- Havana, 1980

Generic Person Specification

Degree or Equivalent, or a Professional or Vocational Qualification in the Subject Area You Wish to Teach: I hold a Bachelor's degree in Mathematics & Statistics with an exceptional GPA of 4.94/5 and a PhD in Computational Mechanics. These qualifications provide a solid foundation in the subject area I wish to teach, ensuring I possess the necessary knowledge and expertise.

An Appreciation of How Adults Learn and an Appreciation of Study Skills: My extensive teaching experience across various universities and institutions has given me a deep understanding of how adults learn. I have developed and delivered numerous courses tailored to adult learners, incorporating study skills and practical applications to enhance their learning experience.

The Ability and Willingness to Promote the Learning of Adults through Correspondence Tuition, Telephone and Face-to-Face Tuition, and, Where Appropriate, Online Tuition: I have taught mathematics to adults through various formats, including face-to-face, correspondence, and online tuition. My role as an Online Tutor Master in Data Science at the Worldwide Network University of London involved conducting engaging webinars and providing timely assistance to students, demonstrating my ability to adapt to different teaching modalities.

The Ability to Use Information and Communication Technology in Teaching and Supporting Students and Communicating with Other Areas: With seven years of experience in Python and cloud environments like Google Cloud Platform, I am proficient in using ICT for teaching and supporting students. I have created interactive computational workflows using Jupyter Notebooks and have utilized various software tools like Matlab, Maple, and Mathematica in my educational projects.

The Ability to Work with Students from Diverse Educational, Cultural and Work Backgrounds: I have taught in various countries, including Cuba, Angola, Argentina, Spain, and Ecuador, working with students from diverse educational, cultural, and work backgrounds. This experience has equipped me with the skills to effectively communicate and connect with students from different walks of life.

The Ability to Work with Students with Disabilities: Throughout my teaching career, I have demonstrated a commitment to inclusivity and accessibility. I have experience working with students with disabilities, ensuring they receive the necessary support and accommodations to succeed in their studies. I have worked at Kingsdale Foundation School (Dec.2022-Present) with students who have learning difficulties in different settings.

A Commitment to Student-Centered Learning: My teaching philosophy revolves around student-centered learning. I design my courses to encourage active participation, critical thinking, and practical application of knowledge, ensuring that students are at the heart of the learning process. I have developed my career didactic projects to improve my skills in this address.

An Understanding of and Commitment to Equal Opportunities Policies and Practices: I am committed to promoting equal opportunities in education. My diverse teaching background has made me sensitive to the needs of all students, and I strive to create an inclusive learning environment where everyone has the opportunity to succeed.

An Organized and Systematic Approach to Work: I have managed multiple research projects and teaching assignments simultaneously, demonstrating my ability to stay organized and systematic in my work approach. This ensures that I can handle the demands of teaching while maintaining high standards of quality and efficiency.

The Potential to Work Successfully in a Team and the Potential to Work Independently: I have collaborated with colleagues on various research projects and teaching initiatives, showing my ability to work effectively in a team. At the same time, I have independently developed and delivered courses, highlighting my capacity to work autonomously when required.

Good Written and Oral Communication Skills: My extensive experience in teaching and research has honed my written and oral communication skills. I have published numerous papers and technical reports and have presented my research at various conferences, ensuring clear and effective communication of complex ideas.

A Commitment to Personal Staff Development: I am dedicated to continuous learning and professional development. Throughout my career, I have actively sought opportunities to expand my knowledge and skills, ensuring that I remain current with advancements in my field and pedagogical practices.

Availability and Accessibility to Students and Ability to Travel to Designated Tutorial Centres: I understand the importance of being available and accessible to students. My previous roles have required flexibility in meeting students' needs, and I am committed to maintaining this availability. Additionally, I am willing and able to travel to designated tutorial centres as required.

By meeting these personal specifications, I am confident in my ability to excel as an Associate Lecturer, providing high-quality education and support to students while contributing positively to the academic community.

Foundation of this application

- **Educational Setting:** I am a passionate applied mathematician focused on numerical computations applied to engineering processes. I hold a Bachelor's degree in Mathematics & Statistics with an outstanding GPA of 4.94/5, demonstrating my strong foundation in analytical and quantitative skills. Additionally, I earned a PhD in Computational Mechanics, where I developed novel numerical methods for coastal hydrodynamics. My research has been extensively published, showcasing my active contribution to the field. Some examples of my research include:
 - Computational Methods for Flooding Risk Management in the Babahoyo River Basin, Ecuador (2016-2018)
 - Numerical Methods in Partial Differential Equations, Mathematics-UNSL, Argentina (2010-2011)
 - Numerical Stabilization of Convection-Diffusion-Reaction Problems, Technical Report 06-03, Delft, Netherlands (2006)
 - Computational Methods for Transport Phenomena in river, estuaries and coasts, UPC, Barcelona, Spain (2002-2008)
 - Development of Basic Resources for Numerical Methods, CIMEC-UNL, Argentina (1998-2002)
- **Teaching Experience** I have successful experience teaching degree-level courses in computational applied mathematics:
- **Courses Taught:**
 - Calculus (2016, 2015, 1982-1986)
 - Numerical Methods, UTB-Ecuador, 2016,
 - Applied Mathematics, UNSL-Argentina, 2013,
 - Ordinary Differential Equations, UPC-Barcelona, Spain (2002, 2005, 2006),
 - Basic Mathematics, ISPEJV-UH-Cuba (1982-1986), UNSL-Argentina, 2000,
 - Modelling and Numerical Methods, UNQ-Argentina, 2000,
 - Series and Partial Differential Equations, ISPEJV-UH-Cuba (1992-1994),
 - Mathematical Analysis, Agostinho Neto University, Luanda, Angola (1986-1987).
 - Calculus, Algebra and Statistics, UOL Worldwide -London (2020-2021)
- **Projects:**
 - *Mathematics and Technology* at Technical University of Babahoyo (UTB), Ecuador (2015-2016): Developed practical learning projects with engineering applications.
 - *Applied Mathematics for Undergraduate Engineers* at National University of San Luis (UNSL), Argentina (2013): Designed a discipline where students solved and presented engineering simulation problems.
- **Mentoring:**
 - Directed the Magister Thesis *The π number and its historical evolution*, Sarmiento University, Buenos Aires, Argentina (2012).
 - Directed the Magister Thesis "Numerical Computations of Irregular Waves in Hyeres" (2004).
 - Directed the Degree Thesis "Waves in Harbors" for a Graduate in Mathematics at Havana University, Cuba (1992).
- **Software Proficiency: The students applied** Matlab, Maple, Mathematica, C++, and Parallel Computing in educational projects.
- **Diverse Teaching Background:** Taught at various institutions including the Ministry of Education, Cuba (1979-1980), Luanda University, Angola (1986-1987), Quilmes University, Argentina (2000), UPC, Barcelona, Spain (2002-2008), and UTB, Babahoyo, Ecuador (2015-2016).
- **Module Concepts and Methods** I have a comprehensive understanding of the main module concepts and methods for the development of this position.

My strong academic content and research experience demonstrate my ability to quickly learn and master new concepts, ensuring I can swiftly develop expertise in any remaining module content.

- **Programming Expertise** I have seven years of experience in Python, working extensively with Jupyter Notebooks and cloud environments like Google Cloud Platform (GCP). During this time, I have developed a strong GitHub profile showcasing projects in Data Science, Statistics, and Numerical Computing. My contributions include:
- **Customer Satisfaction Classification:**

- **Description:** Created a machine learning project to classify customer satisfaction in mixed datasets.
- **Technologies:** Python, R, Deep Learning, Jupyter Notebooks, GCP.
- **Impact:** This model provided highly accurate classifications on large mixed datasets, enabling the company to improve service quality.
- **Tourism-Enhancer-Index for London:**
 - **Description:** Developed a machine learning project to segment tourist activity in London.
 - **Technologies:** Foursquare API, Apache Spark, Python, Geospatial Datasets, Jupyter Notebooks, Deep Learning, Clustering, GCP.
 - **Impact:** Created a recommendation engine that ranked new places for potential touristic activities, aiding city planners and businesses in developing targeted tourism strategies.
- **simleng_ai:**
 - **Description:** Built the Simleng Library, an open-source machine learning package.
 - **Impact:** The SiMLEng package supports education, research, and training by connecting to databases for data modelling strategies, offering a web interface, and generating comprehensive analysis reports. This tool has been widely adopted by educational institutions and research centres for its versatility and ease of use.

Using Jupyter Notebooks, I have created interactive and reproducible computational workflows, such as developing and fine-tuning deep learning models, visualizing results, and documenting processes for collaboration and reproducibility. My experience with cloud platforms like GCP has enhanced my ability to scale computational tasks and collaborate with distributed teams.

Online Teaching Experience From October 2020 to April 2021, I served as an Online Tutor Master in Data Science at the Worldwide Network University of London. During this period, I successfully tutored students in Calculus, Geometry, and Statistics, achieving high student satisfaction and academic success. Key aspects of my role included:

- **Developing and Delivering Lectures:** Conducted engaging webinars on Calculus and Statistics, ensuring complex concepts were accessible to students with diverse backgrounds.
- **Student Support:** Provided timely assistance and constructive feedback through the online platform, directing students to relevant materials to facilitate independent learning.
- **Assessment and Feedback:** Designed and marked quizzes and exams, ensuring evaluations were fair and aligned with learning objectives.

This experience has equipped me with the skills to effectively teach and manage online courses, understanding the challenges and opportunities that online education presents.

REFERENCES

1 Relationship: Ph.D. Thesis Boss	
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2 Relationship: colleague	
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PUBLICATIONS IN DISCIPLINE

- **Author. [Booklet].** Study material of mathematical (Part I) 1981. Published by Rebellious Youth. In Spanish. Cuba.
- **Author. [Text Book].** “Mathematics” (1984). Editorial: People and Education. In Spanish. Cuba.
- **Author. [Book].** Course for primary teachers. Guide to study (1983). In Spanish. Cuba.
- **Author. [Non-INDEXED]** Methods of Solution of non-linear flow equation in an unsaturated aquifer (July 1988).
Memoirs of the XIII Latin American Congress of Hydraulics.
- **Author. [INDEXED]** Mathematics in the profile of the Constructions (article) (January 1989).
Magazine of Hydraulic Engineering. ISSN 0253-0678. Cuba.
- **Author. [INDEXED]** Package of programs “OLAS” (1990).
Journal of hydraulic Engineering. Cuba. Cybernetics. ISSN 0253-0678. Cuba.
- **Author. [INDEXED]** Generation of superficial and interiors wave due to presence of submarine bed.
Journal of Mathematics and Cybernetics (1992). ISSN 0138-8916. Cuba.
- **Authors. [INDEXED]** R. P. Bonet, Nigro N., Storti M. A., “Solution of the equation of Berkhoff with approximates radiation conditions”.
International Journal of Numerical Methods in Engineering Volume 13, 4, 547-557 (1997). ISSN 0213-1315 - in Spanish.
- **Authors. [INDEXED]** R. P. Bonet, Nigro N., Storti M.A., Idelsohn S.R. “Non-local Discrete Absorbing Condition (DNL) in Finite Differences for Elliptic Water Waves Models”, Vol. 14, 4, 481-500, RIMNE (1998). ISSN 0213-1315-in Spanish
- **Authors. [INDEXED]** R.P. Bonet, Nigro N., Storti M.A., Idelsohn S.R. “Non-local Discrete Absorbing Condition (DNL) in Finite Elements for Elliptic Water Waves Models”, Vol. 15, 1, 3-18, RIMNE (1999) , ISSN 0213-1315-in Spanish
- **Authors. [INDEXED]** R. P. Bonet, Nigro N., Storti M. A., Idelsohn S.R. “A Discrete Non-local Absorbing Boundary Condition for External Problems governed by Helmholtz Equation”, International Journal for Numerical Methods in Fluids, Vol. 29, pages 605-621 (1999).
- **Authors. [INDEXED]** R. P. Bonet, Nigro N., Storti M. A., Idelsohn S. R. “A Non-local Discrete (DNL) Outgoing Boundary Condition for Diffraction of Surface Waves; Communications in Numerical Methods in Engineering, Vol.14, pages 849-861 (1998).
- **Authors. [INDEXED]** R. P. Bonet, Nigro N., Storti M.A., Idelsohn S. R. “Non-reflective Planar Boundary Condition based on Gas filter”, IJNME, Vo. 47, pages 969-983 (2000).
- **Aurhors. [Technical Report]** R. P. Bonet, Nigro N., Storti M.A., Idelsohn S. R. “On the stability of β -FEM for the shallow water equations. Technical Report. CIMEC, pages 1-12, (2000).
- **Authors. [INDEXED]** M.A. Storti, J. D’Elia, R.P. Bonet, N.M. Nigro, S.R. Idelsohn “The DNL Absorbing Boundary Condition: Applications to Wave Problems”, Computer Methods in Applied Mechanics and Engineering 182 (3-4) (2000) pages 483-498.
- **Authors. [Technical Report]** R.P. Bonet, N. Nigro, M Storti and S. Idelsohn. “Non-viscid flow in open channels with lateral contraction”. Technical Report, CIMEC-UNL, pages 1-8, (2001).
- **Author. [Technical Report]** R.P. Bonet. “Stabilized Element Methods for Transport Equations”, Technical Report, Barcelona, April 15, (2004).
- **Authors. [INDEXED]** R.P. Bonet, N. Nigro, M Storti and S. Idelsohn. “Strategies to Accelerate the Convergence of the Method DNL Circular”, RIMNE, Vol. 16, 343-358 (2000)
ISSN: 0213-131-in Spanish
- **Author. [Non-INDEXED]** R.P. Bonet. Lectures on Environment “Scattering on Long Waves. Numerical Aspect”, 98-115, Published by CIMNE, (2005).
- **Author. [INDEXED]** R.P. Bonet. “Numerical Solution of Refraction-diffraction of Surface Waves”, RIMNE, Vol. 21 (2), pages 143-168, (2005). ISSN: 0213-1315-in Spanish.
- **Author. [INDEXED]** R.P. Bonet. “Local Absorbent Boundary Conditions to Berkhoff Equitation over a General Shape boundary”, RIMNE, Vol. 21(1), pages 103-133, (2005).
ISSN: 0213-1315- In Spanish.
- **Author. [Technical Report]** R.P. Bonet. “Numerical Stabilization of Convection-diffusion-reaction Problems”, Technical Report 06-03 ISSN 1389-6520, Delft, pages 1-19, (2006).
- **Author. [INDEXED]** R.P. Bonet. “Refraction and Diffraction of Water Waves using finite elements

with a DNL boundary condition". Ocean Eng. Journal ,63, pages 77-89, (2013).

- **Author [INDEXED]** Ruperto P. Bonet A parametric method for the shallow water equations (89-102). Journal of Magazine of the Sciences.

ISSN 2528-8091. Vol1. No 4, October-December 2016

- **Author [Book]** Ruperto P. Bonet. Mathematics I. Editorial Iberoamerican University Foundation. ISBN 978-84-9079-205-6. Jul 2016. 438 pages. Spain-In Spanish.

CONGRESS and CONFERENCES

- **Speaker.** Internal and Surface Waves by the Presence of Underwater Channels. COMAT '93. Matanzas, Cuba. 1993.
- **Speaker.** Asymptotic Methods for the Propagation of Water Waves. IX ENIEF. San Carlos de Bariloche, Argentina, 1995. Published in the proceedings Memory from IX Congress. Authors: R.P. Bonet, Mario A. Storti
- **Speaker.** Solution of Berkhoff Equation with Approximate Radiation Boundary Conditions. MECOM V. San Miguel de Tucumán, Argentina (September 1996). Published in the proceedings Computation Mechanics. Volume XVI, pages 245-254. Editors: Guillermo Este y Bibiana Luccioni.
- **Speaker.** Open Boundary Conditions for Water Waves Propagation Elliptic Models. CIMAFA '97. IV Symposium of Mathematics. Havana, Cuba (24-28 march 1997). (Poster). Not Published. Authors R. P. Bonet, Nigro N., Storti M. A.
- **Speaker.** Discrete Non-Local Absorbing Boundary Condition for Elliptic Models of Water Waves Propagation. X ENIEF. Congress on Numerical Methods and its Applications, ENIEF '97. San Carlos de Bariloche, Argentina (November 1997). Published in the proceedings Computation Mechanics, Volume XVIII, pages 23-32. Editors: E. Dari, C. Padra y R. Silba. Authors: R. P. Bonet, Norberto Nigro, Mario A. Storti, Sergio Idelsohn.
- **Speaker** at keynote lecture. The Discrete Non-Local (DNL) Radiation Boundary Condition. X ENIEF. Congress on Numerical Methods and its Applications, ENIEF '97. San Carlos de Bariloche, Argentina (November 1997). Computational Mechanics. Volume XVIII, pages 143-152. Editors: E. Dari, C. Padra, and R. Silba. Authors: M. Storti, J. D'Ellia, R. P. Bonet and S. Idelsohn.
- **Speaker.** A Discrete Non-Local (DNL) boundary Condition for Elliptic Harbor Wave Models. First International Conference on Applied Engineering to Development of harbor and Marine Waves. Santa Fe, Argentina (June 1998) Not Published. Authors: R. P. Bonet, Norberto Nigro, Mario A. Storti, and Sergio R. Idelsohn.
- **Speaker.** The Discrete Non-Local (DNL) Outgoing Boundary Condition for Water Wave Radiation and Scattering. IV WCCM (World congress). Buenos Aires, Argentina (June 29 - July 2, 1998). Authors: R.P. Bonet, Norberto Nigro, Mario A. Storti, and Sergio R. Idelsohn.
- **Participant.** XII national Congress of Water and Second Symposium of Water Department from South Cone. Santa Fe, Argentina (3-7 August 1998).
- **Speaker.** The Discrete Non-Local DNL sponge Filter as Outgoing Boundary condition for Water Wave Radiation and Scattering. Eighth Chilean Congress of Mechanical Engineering, University of Concepcion, Faculty of Engineering, Department of Mechanical Engineering, Chile (October 27-30, 1998). Published in the proceedings Records Eight Chilean Congress of Mechanical Engineering. Volume II, page 1003-1008) Chile. Authors: R. P. Bonet, Norberto Nigro, Mario A. Storti, and Sergio R. Idelsohn.
- **Speaker.** Finite Elements in Unbounded Domain for Water Waves Refraction-diffraction Problems. PACAM VI (Applied Mechanics in the Americas). Rio de Janeiro, Brazil (January 4-8, 1999). Published in the proceedings Sixth Pan-American of Applied Mechanics and Eight International Conference on Dynamic Problems in Mechanics, volume 6, page 401-404. Authors: R. P. Bonet, Nigro N., Storti M.A., Idelsohn S.
- **Speaker.** A Filtering Layer to Acoustic Waves MECOM 99. Mendoza, Argentina (6-10 September 1999). Published in the proceedings from Sixth Computational Mechanics Congress. Authors R.P. Bonet, Nigro N., Storti A., Idelsohn S.
- **Participant.** Circular DNL Method for Elliptic Harbor Wave Models. XX CILAMCE. Sao Paulo, Brazil (1999). Published in the proceedings from 20th Iberian Latin-American Congress on Computational Methods in Engineering XX CILAMCE, Sao Paulo, Brazil (November 5, 1999). Book: Abstracts edited by Paulo M. Pimienta, Reyolando M.L.R.F. Brazil, Edgard S. Almeida Neto. Authors: R. P. Bonet, Norberto Nigro, Mario A. Storti, and Sergio R. Idelsohn.
- **Participant.** A New Way to Solve Discrete Equations System. ECCOMAS 2000. Barcelona, Spain (September 10-14, 2000). Publication Ref. Number: CMNM 161 (in Conference Proceedings). Authors: R.P. Bonet, Norberto Nigro, Mario A. Storti, Sergio Idelsohn.
- **Speaker.** A Filtering Layer to Absorbing Water Waves. XIX Latino-American Hydraulic Congress. Cordoba, Argentina (October 22-27, 2000). Published in the proceedings. Authors: R. P. Bonet, Norberto Nigro, Mario A. Storti, Sergio Idelsohn.
- **Speaker.** An Explicit Scheme for the Numerical Solution of the Shallow Water Equations. ENIEF2000. Bariloche, Argentina (November 20-24, 2000). Published in the proceedings. Authors: R. P. bonet, Norberto Nigro, Mario A. Storti, Sergio Idelsohn.

- **Participant.** Congress of the Argentinean Mathematical Union. LI Annual Meeting of Sciences Communication. National University of San Luis, Argentina (Sept. 17-21, 2001).
- **Speaker.** A stabilized Finite Element Method for the Shallow Water Equations Using a Finite Increment Calculus Formulation. IN-MAT. Buenos Aires, Argentina (November 7-9, 2001). Published in the proceedings. Authors: R. P. Bonet, Eugenio Oñate.
- **Participant.** Second International Congress on Numerical Methods in Engineering and Applied Sciences. Guanajato, Mexico (January 17-19, 2002).
- **Speaker.** Outgoing Boundary conditions on a Non-circular Open Boundary for Elliptic Harbor Wave Models. VII International Conference on Advances in Turbulence. Barcelona, Spain (21-23 November 2003). Author: R.P. Bonet
- **Speaker.** A comparison of Outgoing Boundary Conditions Strategies for Elliptic Harbor Waves Models. Eccomas04. Jyväskylä Faculty of Engineering, Finland (2004).

Author: R.P. Bonet.

- **Speaker.** An Extended Elliptic Water Waves Model to Simulate Waves in Harbors. Marine 2005. Oslo, Norway (June 27-30, 2005). Published in the proceedings Computational Methods in Marine Engineering. Pages 455-46. ISBN: 84-95999-63-3. Organized by CIMNE/Det Norske Veritas. Author: R. P. Bonet
- **Speaker.** Numerical Simulation of Wave Propagation in Toulon Bay. Waves 2005. Madrid, Spain. (July 3-7, 2005). Published in the proceedings Waves 2005, NIP0163-05-009-006. Organized by CEDX/IACHR. Authors: R.P. Bonet and José María Redondo.
- **Speaker.** Numerical Solution of an Unbounded Wave Guide Problem. IABEM 2006. Graz. Austria (July 10-12, 2006). Published on CD: ISBN: 3-902465-23-9. Organized by Graz University of Technology. Author: R. P. Bonet.
- **Participant.** A domain Decomposition Approach to Computer Wave Refraction-diffraction. ECCOMAS CFD 2006. Egmond aan Zee, The Netherlands, (September 5-8, 2006). Published in the proceedings. Organized by Delft University.
- **Participant.** Real Analysis and Approximation Theory. Congress in memory to Felipe Zó. November 2, 2010. National University of San Luis.
- **Speaker.** Improved discrete non-local absorbing boundary condition for Helmholtz equation. Applications to unbounded Wave Problems. IIIMACI 2011. Vol. 3 pages 267-270. White Bay, Argentina (9-11 May 2011). Organized by University of South. Author: R. P. Bonet.
- **Speaker.** Residual Discrete Non-Local Method for Waves Radiation and Scattering Problems. IV CLAM. Córdoba, Argentina. (August 6-10, 2012). Organized by National University of Córdoba
- **Participant.** First Congress of integration between university and the productive society. October 15-16, 2012. San Luis. Argentina.
- **Participant.** IV MACI 2013. May 2013. Organized by the National Technological University (UTN) Buenos Aires. Argentina.
- **Participant.** Biz Barcelona (May30-31 ,2015) Barcelona. Spain
- **Participant.** Biz Barcelona (May 30-31,2018) Barcelona. Spain